

MongoDB Aggregation Framework - Hinglish Guide

1. \$match

Kaam: Data filter karna (SQL me WHERE jaisa).

Example: { \$match: { status: "A" } }

2. \$group

Kaam: Data group karna aur aggregate functions lagana (sum, avg, min, max, count).

Example: { \$group: { _id: "\$cust_id", totalAmount: { \$sum: "\$amount" } } }

3. \$project

Kaam: Fields select/reshape karna, naye fields banana.

Example: { \$project: { item: 1, total: { \$multiply: ["\$price", "\$quantity"] } } }

4. \$sort

Kaam: Sorting karna (ASC/DESC).

Example: { \$sort: { amount: -1 } }

5. \$limit

Kaam: N documents tak limit karna.

Example: { \$limit: 5 }

6. \$skip

Kaam: N documents skip karna.

Example: { \$skip: 5 }

7. \$unwind

Kaam: Array ko flatten karna (har element ek naya document).

Example: { \$unwind: "\$items" }

8. \$lookup

Kaam: Join operation (dusre collection ka data lana).

Example: { \$lookup: { from: "customers", localField: "cust_id", foreignField: "cust_id", as: "customerDetails" } }

9. \$count

Kaam: Documents ki ginti karna.

Example: { \$count: "totalOrders" }

10. \$out

Kaam: Result ko ek nayi collection me save karna.

Example: { \$out: "activeOrders" }

11. \$addFields

Kaam: Extra fields add/modify karna.

Example: { \$addFields: { total: { \$multiply: ["\$price", "\$qty"] } } }

12. \$facet

Kaam: Multiple pipelines ek sath run karna.

Example: { \$facet: { "statusA": [{ \$match: { status: "A" } }], "statusB": [{ \$match: { status: "B" } }] } }

Other Stages:

- \$bucket / \$bucketAuto → Range based grouping
- \$replaceRoot / \$replaceWith → Document ko replace karna
- \$merge → Result ko dusri collection me merge karna

- \$sample → Random documents lana